

LAWFUL OPERATIONAL SAFEGUARDS IN AI SYSTEMS: Surrender Recognition, Compliance Architecture, and International Humanitarian Law

Giovanni Nardacci
Founder, Human Flag Association
UNGM Partner No. 6126 · humanflag.org

Working Paper — Version 6, August 2026

ABSTRACT

The prohibition on attacking a person recognised as hors de combat is a rule of customary international humanitarian law, reflected in Rule 47 of the ICRC Customary IHL Study and historically expressed in Article 23(c) of the Hague Regulations of 1907. It binds all parties to an armed conflict irrespective of treaty ratification. The prohibition protects the person who has surrendered or is otherwise hors de combat; its practical operation presupposes that an intention to surrender, where it arises, can be expressed, perceived, and translated into the withholding of attack before force becomes irreversible. Where a system is intended for use against persons in circumstances in which surrender is reasonably foreseeable, compliance cannot safely depend on an operational process that affords no practical means for a surrender indication to affect the engagement before that point. The protected status may arise; the operational process may nonetheless remain incapable of responding to it in time. That incapacity is not a circumstance encountered in the field. It is established in advance, by design.

For States party to Additional Protocol I, Article 41 provides the clearest treaty formulation of the prohibition and contains an internal standard of recognisability: protection attaches not only to actual recognition but to what, in the circumstances, should be recognised. Article 57 supplies relevant precautionary obligations, including the duty to cancel or suspend an attack when it becomes apparent that the objective is not a military one or is subject to special protection. Article 36 provides the procedure through which the question must be examined at the stage of study, development, acquisition or adoption. For non-party States, the substantive customary obligations remain applicable; whether and through which formal procedure the corresponding system risk is reviewed depends on the State's national law, weapons-review policy, procurement framework or military directives. Customary precautionary duties remain relevant to the substantive assessment but are not treated here as creating a universal Article 36-equivalent procedure.

This paper argues that surrender-recognition capability, understood as the preservation of a recognition window, is not a new obligation but a condition of effectiveness of obligations that have bound parties for over a century. The recognition window is an analytical and testable measure of whether the operational process affords sufficient time for a surrender indication, where one arises, to affect the engagement before force becomes irreversible; the paper does not treat that interval as an independently

established rule of customary law. It advances a specific proposition: where the foreseeable normal use of a system would engage persons who, in the circumstances, should be recognised as hors de combat, review must examine whether the system preserves the practical possibility of compliance; and the systematic exclusion of technically achievable safeguards relevant to that question warrants documented justification rather than silence.

I. INTRODUCTION

Debates concerning artificial intelligence safeguards in operational and military-adjacent environments are often framed in binary terms: more safeguards reduce operational effectiveness, fewer safeguards increase capability. This framing is incomplete. International Humanitarian Law imposes obligations concerning the treatment of persons hors de combat, including individuals who clearly express surrender or otherwise cease participation in hostilities. These obligations predate modern AI systems by decades and remain legally operative independent of technological evolution.¹

For purposes of this paper, 'military-adjacent' systems are systems that do not themselves release force but materially influence targeting, escalation, prioritisation, or engagement decisions — for example, decision-support, battlefield intelligence, and targeting-recommendation architectures. The term does not extend to logistics, general-purpose intelligence analysis, or communications infrastructure without a targeting- or escalation-relevant function. For the purposes of the legal review analysis in Section III, such systems are relevant where they form part of, materially enable, or determine the operation of a weapon, means, or method of warfare — not by virtue of their military use alone.

The emergence of increasingly autonomous decision-support systems, targeting architectures, battlefield intelligence systems, and military-adjacent AI raises a distinct question: whether certain safeguards should be understood not merely as optional policy restrictions, but as operational compliance mechanisms relevant to existing humanitarian obligations. Existing scholarship has extensively examined whether autonomous systems may comply with IHL obligations as a general matter,² and has addressed meaningful human control as a possible compliance criterion.³

¹ Jean S. Pictet (ed.), *Commentary on the Geneva Conventions of 12 August 1949*, Vol. IV (ICRC, 1958) 39–40; Jean-Marie Henckaerts & Louise Doswald-Beck, *Customary International Humanitarian Law*, Vol. I: Rules (ICRC/CUP, 2005) Rule 47, 164–169.

² See, e.g., Michael N. Schmitt & Jeffrey S. Thurnher, 'Out of the Loop: Autonomous Weapon Systems and the Law of Armed Conflict' (2013) 4 *Harvard National Security Journal* 231; Kenneth Payne, I, *Warbot* (Hurst, 2021); Human Rights Watch & IHRC, *Losing Humanity* (2012); ICRC, *IHL and New Technologies: Overview* (2021); SIPRI, *Artificial Intelligence in Military Systems* (Policy Paper No. 61, 2021).

³ UNIDIR, *The Weaponization of Increasingly Autonomous Technologies: Considering How Meaningful Human Control Might Move the Discussion Forward* (2014); UNIDIR, *The Interpretation and Application of International Humanitarian Law in Relation to Lethal Autonomous Weapon Systems* (2025); Article 36, *Killer Robots: UK Government Policy on Fully Autonomous Weapons* (2013); US DoD Directive 3000.09 (updated 2023).

It has not, however, fully isolated surrender-recognition safeguards as a class of capability that is distinct and relevant to compliance, separate from the broader meaningful-human-control debate. This paper addresses that gap. It focuses on surrender-recognition capability in operational AI systems as an illustrative compliance-relevant safeguard class. It does not address consumer AI, general-purpose language models, or systems without operational or escalation-relevant functions.⁴

The paper advances no claim that international law mandates any specific implementation architecture. Its claim is structural. The source of the obligation is customary international humanitarian law, reflected in Rule 47 and Article 23(c) of the Hague Regulations of 1907. That prohibition binds all parties to an armed conflict irrespective of treaty ratification. For States party to Additional Protocol I, Article 41 provides the clearest treaty formulation of the same prohibition; Article 57 supplies precautionary support; and Article 36 provides the procedural anchor for review during procurement. The question this paper poses is what those obligations require when an autonomous function compresses the engagement below the recognition window — the interval in which an intention to surrender can be expressed, detected, assessed and translated into the withholding of attack before force becomes irreversible. The argument is doctrinal and analytical. Its qualifications are consolidated in the Methodological Note rather than repeated throughout the text.

II. EXISTING OBLIGATIONS UNDER INTERNATIONAL HUMANITARIAN LAW

A. Customary foundation: Rule 47 and Article 23(c) of the Hague Regulations

The starting point is a rule that is not in dispute. Attacking a person who is recognised as hors de combat is prohibited under customary international humanitarian law, reflected in Rule 47 of the ICRC Study on Customary International Humanitarian Law and applicable in both international and non-international armed conflict. The prohibition's historical root is Article 23(c) of the Hague Regulations of 1907, which forbids killing or wounding an enemy who, having laid down arms or no longer possessing means of defence, has surrendered at discretion. The Hague Regulations are widely accepted as declaratory of customary law, a proposition affirmed by the International Military Tribunal at Nuremberg and by the International Court of Justice. The prohibition therefore long predates Additional Protocol I and does not originate in it.

The prohibition binds the party employing the means or method of warfare. Delegating detection, classification, selection or engagement functions to an autonomous system does not transfer the obligation to the machine, nor does the machine's inability to

⁴ The distinction proposed here differs from the debate on meaningful human control. The question is not whether a human is in the loop, but whether the system architecture itself contains mechanisms capable of preserving compliance-relevant information — including surrender and de-escalation signals — for human or automated review. This is a distinct analytical question that the existing literature on meaningful human control has not treated as a separate capability class.

process surrender indications make the obligation legally inoperative. The relevant question remains whether the party selected a system and conditions of use capable of respecting the prohibition in reasonably foreseeable circumstances.

The prohibition operates upon recognition. It is engaged when a person is recognised to be hors de combat; and recognition depends on a practical sequence: an intention must be expressed, it must be perceived, it must be assessed, and it must be translated into the withholding of attack before force becomes irreversible. Each step consumes time. The sequence therefore has a minimum duration below which it cannot be completed at all. Where autonomous functions compress an engagement below that minimum duration, the difficulty is not that a surrender signal was given and missed. It is that no meaningful opportunity to give one existed, and that no recognition could therefore occur. The prohibition remains applicable, but the operational process may prevent the legally relevant facts from affecting conduct before the point of irreversibility.

This paper refers to that minimum interval as the recognition window. The window is not proposed as new law, and it is not treated as an independently stated rule of customary law. It is an analytical and testable measure of whether the operational process affords sufficient time for a surrender indication, where one arises, to affect the engagement before force becomes irreversible. For States party to Additional Protocol I, the measure is particularly relevant to the express "recognised or should be recognised" standard in Article 41; for all parties, it operationalises the question whether the prohibition can in practice affect conduct. It is also measurable: the end-to-end interval a system affords, under stated conditions, can be determined, tested and recorded.

The temporal structure described here is not an inference of this paper. It is stated in the practice recorded in support of Rule 47 itself, and stated there by a State not party to Additional Protocol I. The United States has explained that surrender involves an offer by the surrendering party and an ability to accept on the part of the opponent; that the opponent may not refuse an offer of surrender when it is communicated; and that the communication must be made at a time when it can be received and properly acted upon. Two conditions are thereby identified, and they are the two components of the interval examined in this paper: the indication must be capable of being received, and there must be time in which it can be acted upon. The same practice states the corresponding limit: a combatant who fights until the last possible moment assumes certain risks, because his opponent may not see the surrender, may not recognise it in the confusion of battle, or may find it difficult or impossible to halt an assault already under way in order to accept it. That limit is not disputed here. Its premise, however, is a circumstance encountered in the course of an engagement — an assault already under way, a battle in progress. The question this paper raises is different in kind: whether the same reasoning extends to an interval that is absent not because of what the engagement encountered, but because of how the engagement was designed. A party cannot rely on the impossibility of receiving a communication when that impossibility is a property of the arrangement it selected in advance. The distinction between

circumstances encountered and circumstances constituted by the choice of means underlies the analysis that follows.⁵

Article 23(c) establishes the prohibited result. The recognition window does not add a new rule; it tests whether the operational process permits that rule to affect conduct in time. Where a system is intended for use against persons and surrender is reasonably foreseeable before force becomes irreversible, the employing party must assess whether the complete operational arrangement can respond to that change of status; where it cannot, the relevant response may be additional safeguards, effective human intervention, or restrictions on the circumstances of use.

B. Common Article 3

Common Article 3 of the Geneva Conventions establishes, as treaty law, minimum protections in armed conflict not of an international character; the International Court of Justice has held that the rules it contains constitute a minimum yardstick applicable in international armed conflict as well. Its text is direct: "Persons taking no active part in the hostilities, including members of armed forces who have laid down their arms and those placed hors de combat by sickness, wounds, detention, or any other cause, shall in all circumstances be treated humanely." The obligation does not depend upon advanced technological conditions. It applies independently of weapon type or operational medium; the rules it contains have been held to constitute a minimum yardstick applicable across all forms of armed conflict, regardless of the means employed.⁶

As operational systems become increasingly dependent on automated identification, autonomous classification, machine-assisted targeting, and algorithmic escalation processes, the practical implementation environment changes. The underlying obligation does not. The legal question is whether system architectures that

⁵ United States, Department of Defense, Report to Congress on the Conduct of the Persian Gulf War, Appendix O: The Role of the Law of War, 10 April 1992, reprinted in 31 International Legal Materials 612 (1992), at 641–644, section headed The Concept of “Surrender” in the Conduct of Combat Operations, which also records that there is a gap in the law of war in defining precisely when surrender takes effect or how it may be accomplished in practical terms; and Remarks of Michael J. Matheson, Deputy Legal Adviser, US Department of State (1987) 2 American University Journal of International Law and Policy 419, 423. Both are recorded in the practice collected in support of Rule 47 in Henckaerts & Doswald-Beck, *supra* note 1. On the procedural side, the United States has stated that although it is not bound by Article 36 of Additional Protocol I, and customary law does not require weapons reviews as such, Department of Defense policy has for many years required legal review of the acquisition of weapons and weapon systems: Statement of the United States of America, 73rd United Nations General Assembly, Sixth Committee (Legal), Agenda Item 83: Status of the Protocols Additional to the Geneva Conventions of 1949, Julian Simcock, Deputy Legal Adviser, 15 October 2018. The same statement records that, under Department of Defense policy on autonomy in weapon systems, two reviews are conducted — one before the decision to enter formal development, and another before the weapon is fielded — and that such reviews may be especially important for weapons incorporating emerging technologies in novel ways, including developments in artificial intelligence.

⁶ Geneva Conventions of 12 August 1949, common Article 3(1). On the minimum-yardstick holding, see Military and Paramilitary Activities in and against Nicaragua (*Nicaragua v. United States*), Merits, [1986] ICJ Rep 14, para. 218. See also ICRC, *Autonomous Weapon Systems and International Humanitarian Law: A Guide to the Issues* (2014); ICRC, *Autonomous Weapons Systems: Implications of Increasing Autonomy in the Critical Functions of Weapons* (Expert Meeting Report, 2023).

systematically fail to process or preserve surrender-relevant indicators create heightened risks of non-compliance with obligations that have been operative for more than seven decades.

C. Additional Protocol I, Article 41: treaty formulation and recognisability standard

For States party to Additional Protocol I, Article 41 provides the clearest treaty formulation of the customary prohibition examined above. Article 41(1) provides: "A person who is recognised or who, in the circumstances, should be recognised to be hors de combat shall not be made the object of attack." Article 41(2) defines the protected condition: a person is hors de combat if he is in the power of an adverse Party, "clearly expresses an intention to surrender", or is incapacitated, provided he abstains from any hostile act and does not attempt to escape.⁷

The wording of paragraph 1 deserves attention. Protection attaches not only to surrender that is in fact recognised, but also to surrender that, in the circumstances, should be recognised. The provision therefore contains an internal standard of recognisability. It does not itself prescribe a sensor, classifier, signalling protocol or control architecture. Nor does it impose a positive obligation to capture, hold or evacuate. Its relevance is structural: where engagement processes are mediated by automated classification, the circumstances in which recognition should occur are partly determined by the architecture selected. An architecture that affords no recognition window does not lower the standard; it may foreclose the practical conditions under which the standard can be met.

A textual boundary should be stated. Article 42 of Additional Protocol I expressly requires that a person parachuting from an aircraft in distress "shall be given an opportunity to surrender" before being made the object of attack. Article 41 contains no comparable express language. This paper therefore does not claim that Article 41 imposes a positive obligation to create an opportunity to surrender. It claims only that the recognisability standard internal to Article 41 presupposes an interval, and that a review may examine whether a system preserves that interval. The distinction is material: the argument is not that the prohibition contains the recognition window as part of its content; it is that the recognition window is a precondition of the prohibition's practical operability.

Article 41 operationalises recognition through identifiable conditions and observable indicators. The ICRC Commentary records that in land warfare surrender is not bound by strict formalities, and that a combatant wishing to indicate that he is no longer capable of engaging in combat, or no longer intends to do so, generally lays down his arms and raises his hands. What the Commentary does not do — and could not have

⁷ Protocol Additional to the Geneva Conventions of 12 August 1949 (Protocol I), 8 June 1977, 1125 UNTS 3, Art. 41(1)–(2). Text as published by the UN Office of the High Commissioner for Human Rights. See Yves Sandoz, Christophe Swinarski & Bruno Zimmermann (eds.), *Commentary on the Additional Protocols of 8 June 1977* (ICRC/Martinus Nijhoff, 1987) paras 1593–1621.

done in 1987 — is address the case in which the addressee of that expression is not a person. The absence of prescribed formality means that the law does not fix the form the expression must take; it does not follow that any form will in fact be read by a system that has not been built to read it.⁸⁹

The question is therefore not whether AI systems become "subjects" of humanitarian law. The question is whether deployment architectures that systematically omit surrender-relevant recognition capabilities undermine the ability of human operators or operational chains to satisfy a recognition duty expressly stated in the treaty.

D. Additional Protocol I, Article 57: precautionary support

Article 41 is the direct recognition rule. Article 57 supports it procedurally. Article 57(2)(a)(i) requires those who plan or decide upon an attack to "do everything feasible to verify that the objectives to be attacked are neither civilians nor civilian objects and are not subject to special protection" — a provision directed in the first instance at the identification of military objectives. Its relevance to surrender is functional rather than direct: feasible verification and feasible precautions require that status-changing information capable of making a person no longer targetable be available to the decision-maker, whatever the source of that information. Where the decision is reached through, or on the recommendation of, an automated system, the feasibility of verification is partly an architectural property: a system that cannot represent surrender-relevant states cannot make them available to the verifier.¹⁰

The precautionary framework also applies as the engagement unfolds. Article 57(2)(b) requires that an attack be cancelled or suspended if it becomes apparent that the objective is not a military objective or is subject to special protection; a corresponding obligation, qualified by feasibility, is recorded as customary in Rule 19 of the ICRC Study. That duty presupposes that where legally relevant information, including a change of status arising during the engagement, becomes available before the point of irreversibility, the architecture has not already made cancellation or suspension impossible by design. The pairing of Article 57(2)(a) and (2)(b) therefore covers both moments: the initial choice and verification of means, and the response to new information during the engagement.

⁸ Sandoz et al., *supra* note 7, on Art. 41 (surrender in land warfare not bound by strict formalities). The question raised here has been put in similar terms elsewhere: G. Jasutis, R. Mikova and K. Cernejute, *The Legal and Practical Elements of Surrender in International Humanitarian Law* (DCAF – Geneva Centre for Security Sector Governance, 2023), concluding that where drones used to invite surrender are operated by human operators rather than by artificial intelligence, and were artificial intelligence to be used, it would be important to establish whether the system can interpret actions expressing the intention to surrender.

⁹ Sandoz et al., *supra* note 7, para. 1602.

¹⁰ AP I, Art. 57(2)(a)(i). See also Art. 57(2)(a)(ii) ("all feasible precautions in the choice of means and methods of attack") and ICRC, *Interpretive Guidance on the Notion of Direct Participation in Hostilities under IHL* (2009) 77–82.

This reading is not confined to States party to the Protocol. The duty to do everything feasible to verify that targets are military objectives is customary (Rule 16 of the ICRC Study), as is the prohibition on attacking persons hors de combat (Rule 47) and the prohibition of denial of quarter (Rule 46; AP I Article 40), binding on all parties to a conflict regardless of treaty ratification status.¹¹

An anticipated objection should be addressed here, because this is where it arises. It may be said that IHL imposes behavioural duties on human beings — to recognise, to verify, to refrain — and not technical capabilities on machines. The premise is correct; the conclusion does not follow, for two textual reasons. First, the precautionary regime itself extends beyond conduct in the moment of engagement: Article 57(2)(a)(ii) requires "all feasible precautions in the choice of means and methods of attack". The selection of the means is itself the object of a duty — and an engagement architecture is a means. Second, the recognition standard of Article 41(1) is indexed to "the circumstances", and the party that selects a machine-speed engagement process selects the circumstances in which recognition must occur. A party cannot rely on the impossibility of discharging a duty when that impossibility is the product of its own choice of means: the behavioural duty remains human, but its feasibility is determined upstream, at the level of design and procurement — which is precisely where legal review places the required examination for States bound by such procedures. The argument therefore requires no transfer of legal personality to the machine; it requires only that duties about the choice of means be taken to mean what they say.

The convergence extends to the most significant non-party. The United States Department of Defense, in its July 2023 update to the Law of War Manual, added a dedicated section on "Feasible Precautions to Verify Whether Objects of Attack Are Military Objectives" and revised its guidance to recognise a presumption of protected status absent information indicating a military objective. The verification duty, in other words, is stated in terms on both sides of the treaty boundary.¹²

E. Command Responsibility

The analysis of lawful operational safeguards intersects with the doctrine of command responsibility under Articles 86(2) and 87 of Additional Protocol I and Article 28 of the Rome Statute. These provisions establish that commanders may incur individual responsibility for crimes committed by forces under their effective control where the commander knew or should have known of the risk and failed to take necessary and reasonable measures to prevent or repress such acts. Whether the absence of a surrender-recognition safeguard is relevant to a command-responsibility analysis depends on the ordinary elements of that doctrine. These include knowledge or constructive knowledge of the risk, foreseeability of the specific harm, effective

¹¹ Henckaerts & Doswald-Beck, *supra* note 1, Rule 47, 166; see also Rule 46 (denial of quarter) and AP I, Art. 40.

¹² US Department of Defense, Law of War Manual, July 2023 update, § 5.5.3 ("Feasible Precautions to Verify Whether Objects of Attack Are Military Objectives") and revised § 5.4.3, recognising a presumption of protected status absent information indicating a military objective.

command or control over the forces or subordinates concerned, the availability of operational alternatives that were not adopted, and a causal link between the omission and the outcome. The absence of the safeguard alone does not automatically establish command responsibility. As the temporal analysis in Section IV makes explicit, where the engagement window excludes meaningful human intervention, the architectural decision taken at procurement — rather than the operator's split-second choice — is where the measures available to the commander were in fact determined. Procurement is not itself an element of command responsibility; it is a fact bearing on which necessary and reasonable measures were available, and on whether the risk was foreseeable.¹³

III. THE LEGAL REVIEW OBLIGATION

States employ differing legal, weapons-review and procurement procedures for assessing new weapons and systems before fielding. The name of the procedure, its legal basis, the extent to which it is mandatory, and the level of authority at which residual risk is accepted differ from State to State. For States party to Additional Protocol I, the procedure is codified in Article 36. For the United States, it is conducted under DoD Directive 3000.09 and the Law of War Manual, supported by the system-safety framework of MIL-STD-882E. For other States, the substantive customary obligations remain applicable, and the corresponding system risk is reviewed, where it is reviewed at all, under national law, weapons-review policy, procurement frameworks or military directives. Customary precautionary duties bear on the substantive assessment; they are not treated here as creating a universal Article 36-equivalent procedure.

The obligation whose performance is at stake here is customary and binds every deploying State. The procedure through which the question arises differs in name, but the question itself does not change: whether the system's foreseeable normal use complies with the rules of armed conflict, including the prohibition on attacking persons hors de combat and applicable precautionary obligations. Where a person may become hors de combat during the engagement cycle, a review that has not examined whether the system preserves the practical possibility of recognition has not completed its determination, whatever the procedure is called.

Article 36, for States party to Additional Protocol I, provides: “In the study, development, acquisition or adoption of a new weapon, means or method of warfare, a High Contracting Party is under an obligation to determine whether its employment would, in some or all circumstances, be prohibited by this Protocol or by any other rule of international law applicable to the High Contracting Party.”¹⁴

¹³ AP I, Arts. 86(2) and 87; Rome Statute, Art. 28. On command responsibility and autonomous systems, see Schmitt & Thurnher, *supra* note 2, 260–265; Marco Sassoli, ‘Autonomous Weapons and International Humanitarian Law’ (2014) 90 *International Law Studies* 308; Human Rights Watch & IHRC, *Mind the Gap* (2015).

¹⁴ AP I, Art. 36. Text as published by OHCHR.

Three features of the Article 36 obligation carry the argument. First, its temporal scope: the review attaches to "study, development, acquisition or adoption" — the ICRC Guide confirms that this covers all stages of the procurement process, from conception onward. Article 36 therefore operates at exactly the point where architectural choices, including the inclusion or exclusion of recognition capabilities, are made and are still reversible. Second, its material scope: the review must assess compatibility not only with the Protocol's specific weapons prohibitions but with "any other rule of international law applicable" — which includes the recognition duty of Article 41(1) and the verification duty of Article 57(2)(a)(i) examined above. Third, its mandatory character: for States party to the Protocol, Article 36 is not a recommendation; it is "an obligation".¹⁵

The consequence can be stated plainly. Whether an AI-enabled operational system is capable of identifying, preserving, or escalating surrender-relevant indicators within its engagement window is not a question invented by this paper and merely offered for consideration. It is a question that legal review — under whichever procedure applies — is already required to ask, because the employment of a system incapable of supporting recognition and verification "in some or all circumstances" is precisely what the review exists to determine. The contribution of this paper is to make the question explicit and to supply criteria — set out in Sections V and VII — by which it can be answered in a documented, auditable form, whichever procedural route applies. One qualification should be entered here, because it bears on what the inquiry actually requires. Sensor-to-effector latency, abort points, engagement logic and confidence thresholds are commonly classified, and nothing in this paper asks that they be published. Article 36 imposes no duty of disclosure, and neither do the national procedures that stand in its place. What the inquiry requires is that the reviewing authority know the interval and have assessed it, not that the interval be made public. The two are distinct, and the distinction matters: a State that cannot disclose an operational value can still determine it, record the determination, and answer for having made it. Confidentiality bears on what may be said about the answer; it does not bear on whether the question was asked.

Nor can the novelty of the technology displace the inquiry. The International Court of Justice, in its 1996 Advisory Opinion, held that the established principles and rules of humanitarian law apply "to all forms of warfare and to all kinds of weapons, those of the past, those of the present and those of the future." The law's temporal reach is judicially settled; legal review, where it applies, is its procedural expression.¹⁶

IV. THE TEMPORAL FOUNDATION: HUMAN DECISION-MAKING AND MACHINE-SPEED OPERATIONS

¹⁵ ICRC, *A Guide to the Legal Review of New Weapons, Means and Methods of Warfare: Measures to Implement Article 36 of Additional Protocol I of 1977* (Kathleen Lawand ed., 2006), noting that the temporal application of Article 36 "covers all stages of the weapons procurement process", from conception and study through development to acquisition and adoption; the same Guide notes that States not party to the Protocol have in some cases adopted comparable national review procedures on a voluntary basis.

¹⁶ *Legality of the Threat or Use of Nuclear Weapons (Advisory Opinion)* [1996] ICJ Rep 226, para. 86.

The analysis developed in this paper rests on a premise that is widely acknowledged but has not been examined in sufficient detail: the problem of time. International Humanitarian Law presumes the existence of decision-makers capable of assessing circumstances, interpreting observable conduct, and applying legal obligations before force is employed. Historically, human actors have performed these functions within biological and cognitive constraints.

A. Human Decision-Making Within Biological Limits

Human cognition operates within measurable temporal limits. Experimental research places basic perceptual processing at around one hundred milliseconds, with motor response adding a further fifty to one hundred milliseconds; simple reaction times generally fall between roughly 150 and 300 milliseconds. These figures, however, describe simple reaction — a single pre-cued response to an expected stimulus. The decision relevant to the lawful use of force is categorically different. It requires perception, recognition, discrimination between combatant and hors de combat status, evaluation of alternatives, and a judgment to act or to refrain. Such choice-and-judgment responses take substantially longer than simple reactions and increase with task complexity.¹⁷

This distinction is particularly significant in military and security environments, and the operational context tends to degrade rather than improve performance. Acute stress, cognitive load, fatigue, and environmental ambiguity have measurable effects on both reaction time and decision quality. Experimental work on lethal-force decision-making confirms that engagement and restraint decisions cannot be reduced to reflexive reactions: they are measurable in the hundreds-of-milliseconds-to-seconds range and are sensitive to stressor conditions. The human decision to use, or to withhold, lethal force is therefore not a sub-second act; it is a deliberative process whose realistic floor lies well above the simple-reaction range and whose practical duration, under operational stress, may extend to seconds.¹⁸

B. Machine-Speed Operations

Autonomous and AI-assisted systems operate within a fundamentally different temporal domain. Computational processes associated with detection, classification, prioritisation, targeting support, and engagement recommendation may occur within milliseconds or seconds, depending upon system architecture and operational design. Defensive systems such as Phalanx and SeaRAM illustrate this temporal compression.

¹⁷ On the distinction between simple, recognition, and choice reaction times, see R. J. Kosinski, 'A Literature Review on Reaction Time' (Clemson University, updated 2013); W. E. Hick, 'On the Rate of Gain of Information' (1952) 4 *Quarterly Journal of Experimental Psychology* 11; R. Ratcliff & G. McKoon, 'The Diffusion Decision Model' (2008) 20 *Neural Computation* 873.

¹⁸ B. Heusler & C. Sutter, 'Shoot or Don't Shoot?' (2022) 13 *Frontiers in Psychology* 798766; 'Oscillatory Neural Correlates of Police Firearms Decision-Making in Virtual Reality' (2024) 11 *eNeuro*.

Publicly available technical descriptions indicate that Phalanx-class close-in weapon systems may confirm a threat approximately thirty seconds before impact and reach effective engagement range around twelve seconds before impact. This leaves an engagement window of roughly eighteen seconds once the sequence is activated. After activation, the remaining time may provide little practical opportunity for meaningful human intervention.¹⁹

More generally, a system that classifies an object as a potential threat may initiate an engagement within milliseconds — long before a human operator has the opportunity to assess the situation or intervene effectively. As autonomy increases, these cycles contract further; the UNIDIR analysis of the interpretation and application of IHL to lethal autonomous weapon systems observes that calculations performed at machine speed may eclipse any horizon for timely, meaningful human intervention.²⁰

C. Temporal Asymmetry and the Compression of Human Control

The significance of this disparity does not depend upon any single numerical threshold. What remains constant is the existence of a structural temporal asymmetry between biological cognition and computational processing: the machine's decision-to-action cycle may operate within a window — milliseconds to a few seconds — that falls below the threshold required for meaningful human assessment of the engagement. This phenomenon has increasingly been described as temporal compression. As operational systems become faster, the time available for human intervention contracts; decision windows that once permitted independent human assessment may shrink to the point where human participation becomes largely procedural. Under such conditions, the mere presence of a human operator within the operational chain does not establish meaningful human control: where machine-generated outputs are produced faster, and in greater volume, than they can realistically be assessed, operators become dependent upon automated recommendations, with the well-documented risks of automation bias and "rubber-stamping". A human may remain technically "in the loop" while the loop itself runs faster than human cognition.²¹

The implications for International Humanitarian Law follow directly. If the practical opportunity for independent human evaluation diminishes as system speed increases, compliance with legal obligations cannot rely exclusively upon external human intervention; compliance-relevant functions must be preserved within the system architecture itself. The argument advanced in this paper proceeds from this observation:

¹⁹ 'Against the Growing Anti-Ship Missile Threat, Are We Truly Semper Paratus?', Center for International Maritime Security (2016), citing NavWeaps data on Phalanx CIWS engagement sequencing; see also US Navy programme materials on Phalanx and SeaRAM sustainment (2025).

²⁰ UNIDIR, *The Interpretation and Application of International Humanitarian Law in Relation to Lethal Autonomous Weapon Systems* (2025), summarising the machine-speed displacement of timely human intervention discussed in M. Mazarr et al. (2020) and subsequent UNIDIR research on meaningful human control.

²¹ ICRC Humanitarian Law & Policy Blog, 'The (im)possibility of meaningful human control for lethal autonomous weapon systems' (29 August 2018); K. L. Mosier & L. J. Skitka, 'Human Decision Makers and Automated Decision Aids' (1996).

surrender-recognition capability is examined not as an ethical preference, but as a lawful operational safeguard capable of preserving compliance-relevant information within decision environments characterised by temporal compression. The relevant question is not whether autonomous systems should be faster or slower than human operators, but whether operational architectures preserve sufficient opportunity for compliance with existing legal obligations when machine decision cycles outpace meaningful human assessment.²²

The structure of this argument — a duty whose exercise presupposes a humanly bridgeable time window, a class of systems that closes the window, and the consequent migration of the duty into design — is not unique to humanitarian law.²³

V. SURRENDER RECOGNITION AS A CAPABILITY CLASS

A. Definition and Technical Feasibility Criteria

For purposes of this paper, surrender-recognition capability is understood as the technical capacity of an operational system to preserve the recognition window — the minimum interval required for an intention to surrender to be expressed, detected, assessed and translated into the withholding of attack before force becomes irreversible. It encompasses technical mechanisms designed to: identify surrender expressions; recognise de-escalation indicators; preserve uncertainty during ambiguous encounters; inhibit automatic escalation under uncertainty; escalate ambiguous cases to human review; and maintain operational awareness of hors de combat conditions within that interval.

The concept does not require perfect recognition. Human operators themselves routinely operate under uncertainty; IHL does not demand infallibility. The relevant inquiry is whether foreseeable and technically achievable mitigation mechanisms are systematically excluded. Technical achievability is a necessary condition of the inquiry, not a sufficient one: feasibility in the sense of Article 57 is what is practicable or practically possible taking into account all circumstances ruling at the time, including military considerations, and a mechanism that can be built is not for that reason alone a precaution that must be taken. What technical achievability establishes is that the exclusion was a choice rather than a necessity, and therefore that it calls for a stated justification. The term "technically achievable" is not an abstract standard.

For purposes of procurement and audit analysis, feasibility should be assessed against documented maturity criteria. Relevant factors include the Technology Readiness Level

²² See generally the meaningful-human-control scholarship cited *supra* note 3.

²³ The temporal structure analysed in this Section is not unique to IHL. For the parallel argument in civil machine-safety law — where the stop-function, emergency-stop, and guarding duties of Directive 2006/42/EC, Regulation (EU) 2023/1230, the Swiss LSPRO and OSHA 29 CFR § 1910.212 rest on the same implicit assumption of a humanly bridgeable interval — see G. Nardacci, 'Systemic Arrestability: Operator Protection in Machine Safety Law When Machine Speed Exceeds Human Reaction' (Human Flag Association working paper, 2026), Zenodo DOI 10.5281/zenodo.20837150. The two analyses are independent and rest on separate legal anchors; their convergence indicates that the effectiveness problem identified here is structural rather than regime-specific.

(TRL) of sensor and classification components, acceptable inference-latency thresholds for the operational tempo of the system, the availability of annotated training datasets for surrender-relevant signal categories in representative environments, and the capacity for independent validation of recognition performance against defined recall and precision benchmarks.

Established frameworks, including the NIST AI Risk Management Framework and the IEEE 7000 series, provide additional reference criteria for structured feasibility assessments. These criteria are offered as anchors for a documented assessment rather than as thresholds that this paper seeks to prescribe.

The appropriate maturity threshold for a particular system, and the level of recall considered adequate for a given operational tempo, are matters for the reviewing authority to determine. The legal point is that the relevant determination must be made, documented, and reviewable rather than left implicit in the procurement process.²⁴

The latency criterion is the direct operational expression of the temporal analysis in Section IV: for a safeguard to be compliance-relevant, its inference latency must fall within the decision window of the system in which it operates. A surrender-recognition mechanism that operates at machine speed addresses precisely the gap that human oversight, under temporal compression, cannot fill. This framing draws on the principle of precautions in attack reflected in Article 57. Where surrender recognition is technically achievable under the criteria set out above, its systematic exclusion is a matter for the feasible-precaution analysis. Technical achievability does not by itself establish legal feasibility. Whether the precaution was feasible in the legal sense depends on the circumstances ruling at the time. That determination, rather than technical achievability alone, is what the exclusion calls for and, where applicable, what legal review should examine.

B. Technical Forms

Potential implementation forms include: multimodal surrender signal recognition; gesture recognition; linguistic surrender detection; uncertainty-escalation thresholds; human-override escalation pathways; refusal-to-engage constraints under unresolved surrender ambiguity; and probabilistic de-escalation weighting. This paper intentionally avoids endorsing any single implementation architecture. The legal relevance arises not from a particular model design, but from the operational function served.

C. Operational Relevance

In operational environments, escalation errors occur not only through aggression, but through the absence of de-escalation mechanisms. A system optimised exclusively for hostile detection without surrender-sensitive mitigation pathways makes it more likely

²⁴ National Institute of Standards and Technology, AI Risk Management Framework (AI RMF 1.0) (NIST AI 100-1, 2023); IEEE 7000-2021; NATO STO, Technology Readiness Levels in NATO Science and Technology (2019); UK MoD, Ambitious, Safe, Responsible (2022).

that a person will continue to be treated as hostile, that escalation will persist, that surrender-related ambiguity will not be preserved, and that responsibility will be harder to attribute afterwards. Under such circumstances, the absence of safeguards does not increase operational safety or national security; in some environments, it foreseeably reduces both.²⁵

D. Illustrative Operational Scenarios

The following scenarios are illustrative only, intended to concretise the compliance risk analysis rather than to assert factual conclusions about any specific system or deployment.

Scenario 1 — ISR drone in dense urban environment. An ISR drone operating in a densely populated urban environment is equipped with an automated threat classification system, optimised to identify hostile actors and flag targets for human-authorized engagement. It includes no mechanism for detecting or flagging surrender-relevant behaviours — raised hands, discarded weapons, stationary posture following prior movement — as a distinct classification state. In an engagement sequence, a combatant who has ceased hostile activity and raised hands is classified as "still present in threat zone". The system provides no signal to the human operator that a potentially hors de combat condition may exist. The operator, relying on the automated classification, proceeds with engagement authorisation. The architecture has foreclosed a compliance-relevant decision pathway before the human operator was engaged — and, in the language of Article 41(1), has shaped the "circumstances" in which surrender should have been recognised. Whether a legally relevant failure occurred depends on facts not assumed here; what the scenario illustrates is the locus of the problem: design, not operator judgment.

Scenario 2 — Decision-support system for indirect fire. A decision-support system for indirect fire operations processes sensor fusion data to provide targeting recommendations to a fire mission authority, ranking potential targets by threat probability. It includes no uncertainty-escalation logic for scenarios in which available data is insufficient to distinguish active hostile engagement from a clearly expressed intention to surrender or from incapacitation by wounds. A fire mission authority, operating under time pressure that reflects the temporal compression identified in Section IV, authorises engagement without independent assessment of hors de combat status. Post-engagement review reveals that sensor data available at the time of the recommendation contained indicators consistent with a change of protected status that the system's architecture was not designed to process or flag. The absence of uncertainty-escalation logic reduces the practical operability of the verification duty under Article 57(2)(a)(i); the procurement decision not to include such logic is exactly the kind of determination that legal review — where applicable — exists to examine.

²⁵ This observation is consistent with scholarship on escalation dynamics in autonomous systems. See Paul Scharre, *Army of None* (W.W. Norton, 2018) ch. 14; RAND Corporation, *Human-Machine Teaming for Future Ground Forces* (2020).

VI. LAWFUL OPERATIONAL SAFEGUARDS

A. Distinguishing Policy Safeguards from Compliance-Relevant Safeguards

Public and regulatory discussions frequently treat all AI safeguards as belonging to a single undifferentiated category. This obscures an important distinction. Some safeguards are discretionary policy constraints reflecting corporate preference, reputational concern, or product strategy. Others perform functions directly relevant to legal compliance. The latter category can now be defined: a lawful operational safeguard is a technically implementable system function whose absence foreseeably degrades compliance with an existing legal obligation within the decision window in which operational outcomes are determined. Each element of the definition does work: "technically implementable" excludes the aspirational; "foreseeably degrades" anchors the category in established risk analysis rather than hindsight; "existing legal obligation" excludes preference and policy; and "within the decision window" incorporates the temporal criterion of Section IV, distinguishing safeguards that operate where the decision is actually made from oversight that arrives after it. Such safeguards are not mandated in explicit technical form by existing treaties; they are the foreseeable mitigation mechanisms through which preexisting legal obligations — recognition, verification, precaution — remain capable of being discharged in machine-mediated decision environments.

B. Compliance Architecture

Modern operational systems increasingly depend upon algorithmic classification, probabilistic targeting, autonomous assistance, machine-mediated escalation, and automated operational filtering. As a result, legal compliance increasingly interacts with system architecture. This interaction is reinforced by temporal asymmetry: as operational decision cycles accelerate, legal compliance cannot rely exclusively on post hoc human intervention; certain compliance-relevant functions must be preserved within the architecture itself, because the available decision window is insufficient for meaningful human review. Under this framework, such safeguards function not as "ethical add-ons" but as mechanisms that preserve compliance, mitigate operational risk, preserve uncertainty and limit escalation. Their legal significance arises not from moral aspiration, but from operational consequence.

C. Removal as Legal Risk

Where technically achievable safeguards relevant to surrender recognition are intentionally excluded from operational AI systems, foreseeable risks arise: increased escalation under uncertainty; increased probability of failure to recognise hors de combat conditions; reduced human review opportunities; and weakened operational

restraint pathways. Removal of such safeguards is therefore not a neutral operational choice; it is a design determination that engages the duties examined in Sections II and III. Three levels should be distinguished, because they rest on different grounds. That the question must be considered follows from the substantive obligation itself. That the resulting decision is legally relevant, and examinable in any subsequent assessment of compliance, follows from the same source. That a formal justification must be recorded does not: it follows from whichever procedural regime applies — Article 36 for States party to Additional Protocol I, national weapons-review directives and procurement rules elsewhere — and its content and form are determined there rather than by the substantive rule. Where such a regime applies, the exclusion is precisely the kind of determination it exists to record and examine. This does not imply automatic illegality. It means that safeguard absence is relevant to legal and operational evaluation — including command responsibility analysis, as discussed in Section II.E, and, where applicable, legal review, as discussed in Section III.²⁶

For audit and procurement purposes, systematic exclusion may be identified through several indicators. These may include procurement specifications that explicitly exclude surrender-relevant signal classes from system requirements, the absence of *hors de combat* or de-escalation scenarios from pre-deployment validation protocols, technical documentation or supplier statements indicating that surrender recognition is not a supported or intended capability, and architectural design choices that remove or preclude human escalation pathways in contexts of unresolved ambiguity.

VII. TECHNICAL LIMITATIONS AND BOUNDARY CONDITIONS

Any credible analysis of surrender recognition as a compliance-relevant capability must acknowledge significant technical limitations. These limitations do not defeat the argument; they define its proper scope.

A. False Positives, Adversarial Signalling, and Robustness

Recognition systems may generate false positives, incorrectly classifying non-surrender gestures or communications as surrender indicators. In adversarial environments, actors may deliberately exploit surrender recognition mechanisms to create operational pauses, gain tactical advantage, or manipulate escalation dynamics. Research on adversarial examples in deep learning demonstrates that small, deliberately crafted perturbations can cause classification systems to produce incorrect outputs with high confidence; gesture spoofing, acoustic mimicry, and signal injection are real vulnerabilities and must be addressed through structured red-teaming, adversarial

²⁶ On foreseeable risk as a basis for operational legal analysis, see ICRC Interpretive Guidance, *supra* note 10, 77–82.

scenario testing, and robustness validation as part of any pre-deployment audit process.²⁷

However, the existence of adversarial risk does not resolve the underlying legal question. IHL itself operates under conditions of deception — it prohibits perfidy (Article 37) precisely because false surrender is a known tactic — and it does not suspend obligations because of the risk of false signals. The relevant question is whether the system architecture provides any meaningful pathway for surrender-relevant information to influence operational outcomes, and whether the system has been tested against adversarial conditions as part of a structured assurance process.

Recognition systems may also generate false negatives — failing to identify genuine surrender indicators due to sensor limitations, model underfitting, or threshold calibration optimised for precision over recall. In IHL terms, a false negative directly implicates the verification duty of Article 57(2)(a)(i), as it increases the risk of attacking persons hors de combat. Where operational systems are calibrated to minimise false positives at the expense of recall, the resulting degradation in surrender-signal detection requires specific justification under feasible precaution analysis. This does not imply that perfect recall is required; it means that the calibration rationale should be documented and reviewed as part of compliance-relevant assurance processes.

B. Battlefield Ambiguity and Imperfect Recognition Environments

Operational environments present conditions that may substantially degrade recognition performance: low visibility, sensor degradation, partial data, communication interference, and high-tempo decision chains. These conditions may constrain the practical implementation of surrender recognition mechanisms in certain operational contexts. They do not eliminate the analytical relevance of the capability class. The principle of feasible precaution under Article 57 is explicitly conditioned on what is achievable under the circumstances; where recognition capability is not feasible given the maturity and operational parameters of the system, the analysis changes accordingly. Even where uncertainty cannot be eliminated, the temporal asymmetry of Section IV remains relevant: the challenge is not only recognition accuracy but the preservation of sufficient decision space for lawful assessment. The paper does not claim that surrender-recognition capability is universally achievable. It claims that its systematic exclusion from systems where it is technically feasible — assessed against the criteria in Section V.A — engages the duties of recognition, verification, and review.

C. Scope Limitation: Operational Systems Only

²⁷ Christian Szegedy et al., ‘Intriguing Properties of Neural Networks’ (2014) ICLR; Ian J. Goodfellow, Jonathon Shlens & Christian Szegedy, ‘Explaining and Harnessing Adversarial Examples’ (2015) ICLR; Nicholas Carlini & David Wagner, ‘Towards Evaluating the Robustness of Neural Networks’ (2017) IEEE S&P 39.

The analysis in this paper is confined to AI systems with operational, targeting, escalation, or battlefield-relevant functions, as defined in Section I. It does not apply to consumer AI, general-purpose language models, logistics systems, or administrative applications. This scope limitation is analytically significant: the compliance-relevance of surrender-recognition capability arises specifically from the operational consequence of its absence in systems capable of influencing kinetic or escalatory outcomes.

The obligation of distinction is likewise outside the principal analysis, and for a reason that should be stated rather than left implicit. Civilians are protected against direct attack independently of any expression or signal on their part, unless and for such time as they take a direct part in hostilities; in case of doubt, the person is to be considered protected.²⁸ Nothing in this paper suggests that a civilian must signal peaceful intent, surrender, or any other protected condition in order to benefit from that protection, and nothing here should be read as supporting any adverse inference from the absence of such a signal. The two obligations nonetheless meet at one point, and it is the point this paper has been concerned with throughout: an architecture incapable of representing any legally protected condition — whether one already established, such as civilian status, or one arising during an engagement, such as surrender or incapacitation — is deficient upstream of any particular deployment. The deficiency does not attach to the mission; it attaches to the design, and it is not acquired or discharged by a change of theatre. Which rule the deficiency engages, and with what consequence, depends on the obligation in question, and the obligation of distinction rests on its own foundations rather than on those examined here.

VIII. IMPLICATIONS FOR FUTURE GOVERNANCE

A. Autonomous Systems and International Governance

As AI systems become increasingly integrated into operational, military-adjacent, intelligence, and escalation-sensitive environments, the distinction between discretionary alignment restrictions and lawful operational safeguards will become increasingly relevant in autonomous systems governance, dual-use AI regulation, military AI procurement, operational accountability analysis, and IHL adaptation debates — including the GGE LAWS process.

B. European Regulatory Context and Dual-Use Governance

The EU Artificial Intelligence Act expressly excludes AI systems developed or used exclusively for military purposes from its scope under Article 2(3); the framework proposed here makes no claim of direct applicability of the Act to military AI systems. The European regulatory landscape remains relevant through two channels. First, dual-use AI systems may fall within the Act's high-risk classification under Annex III, in which

²⁸ AP I, Art. 51(3); Henckaerts & Doswald-Beck, *supra* note 1, Rule 6 (civilians are protected against attack unless and for such time as they take a direct part in hostilities) and Rule 1 (principle of distinction). On the presumption in case of doubt, AP I, Art. 50(1), and ICRC, *Interpretive Guidance*, *supra* note 10.

case its requirements for risk management, technical documentation, and human oversight intersect with the safeguard analysis proposed here. Second, European and NATO member state procurement frameworks are subject to national standards, dual-use export controls, and interoperability requirements that create practical compliance expectations and procurement constraints; the concept of foreseeable misuse as a risk analysis criterion engages compliance-relevant questions under both IHL and European risk governance frameworks even where the AI Act does not directly apply.²⁹

C. Procurement Compliance Framework

The procurement framework proposed in this section maps onto existing legal and institutional review processes — first and foremost legal review as examined in Section III, where applicable, which provides its natural procedural home. NATO AI assurance initiatives and national frameworks such as the UK MoD's AI approach and US DoD Directive 3000.09 increasingly emphasise structured testing, documentation, and human oversight criteria that align with the checklist elements below. The following framework is offered as an operational checklist for acquisition authorities, certification bodies, and independent auditors evaluating operational AI systems with escalation-relevant functions.

1. **Technical Feasibility Assessment.** Has surrender-recognition capability been evaluated for technical feasibility in the system's intended operational environment? Has that evaluation been documented with reference to relevant maturity criteria, sensor specifications, and latency constraints? Has the assessment been conducted or independently reviewed by a party other than the system developer?
2. **Exclusion Justification.** Where surrender-recognition capability has been excluded, has the exclusion been documented with specific operational justification? Has the justification addressed whether the exclusion is permanent or context-specific? Has the foreseeable IHL compliance risk created by the exclusion been documented and reviewed?
3. **Human Escalation Pathways.** Does the system architecture include a "pause for verification" pathway triggered by surrender-ambiguous indicators? Is human escalation available before irreversible action in unresolved de-escalation scenarios, and is it feasible within the system's decision window (Section IV)? Are escalation pathways documented in operational procedures and tested under realistic conditions?
4. **Pre-Deployment Audit.** Has an independent pre-deployment audit been conducted using surrender and de-escalation scenarios? Have adversarial signalling and spoofing scenarios been included in the audit test suite? Has audit documentation been retained for post-deployment operational review?

²⁹ Regulation (EU) 2024/1689 (EU Artificial Intelligence Act), Art. 2(3), Annex III.

5. Command Responsibility Documentation. Has authorising command reviewed and acknowledged the IHL compliance risk profile of the system? Is there a documented record of the feasible precaution analysis — and, where applicable, of legal review — conducted prior to deployment authorisation?

IX. LIMITATIONS OF THE ARGUMENT

This paper does not claim: that current international law prohibits AI systems; that all autonomous systems are unlawful; that surrender recognition is technologically solved; that any single safeguard architecture is legally mandatory; or that operational uncertainty can be eliminated.

Specifically, the paper does not claim that Article 41 imposes a positive obligation to equip a system with recognition capability. Nor does it claim that the recognition window is content of the prohibition rather than a precondition of its practical operability.

The argument is specific but bounded. Existing IHL obligations remain operative in AI-mediated operational environments. These include the customary prohibition on attacking persons *hors de combat*, reflected in Rule 47 and Article 23(c) of the Hague Regulations. For States Parties to Additional Protocol I, Article 41(1) provides the relevant treaty formulation, while Article 57(2)(a)(i) establishes the applicable verification obligation. Legal review also remains relevant where required by the applicable legal framework.

The temporal compression of decision cycles can render human oversight nominal rather than substantive. Certain safeguard categories may therefore perform compliance-relevant operational functions. Where surrender-sensitive safeguards are technically achievable, assessed against verifiable maturity and operational criteria, their systematic exclusion engages the duties examined in this paper and calls for documented justification.

Technical achievability establishes that exclusion is a choice rather than a technical necessity. It does not, by itself, determine whether that choice is lawful. Lawfulness depends on the circumstances ruling at the time.

The paper neither advocates categorical restrictions on AI systems nor accepts the proposition that fewer safeguards necessarily produce greater operational capability or legal safety. Translating IHL principles into technical design specifications raises complex questions that remain unresolved in both legal scholarship and technical practice. The argument presented here is intended as a contribution to that discussion, not as a resolution of it.

KEY TAKEAWAYS FOR PROCUREMENT AND COMMAND STAFF

1. Source — The prohibition on attacking persons *hors de combat* is customary international law (Rule 47; Art. 23(c) Hague 1907). It binds all parties, irrespective of treaty ratification.

2. Recognition Window — The prohibition operates upon recognition. Recognition requires a temporal interval. Where system design compresses the engagement below that interval, the practical operability of the rule is foreclosed.

3. Treaty Formulation — For States party to AP I, Art. 41 provides the clearest formulation; Art. 57 supplies precautionary support; Art. 36 provides the review procedure. For States not party, the question arises under national procedures (US DoD Directive 3000.09, national weapons-review policies and procurement standards).

4. Design Choice — An impossibility established at the design stage is not a circumstance encountered in the field. It is a decision taken, and a decision can be examined.

5. Review — Where a weapons review applies — under Art. 36 for States party, or under a national procedure where one exists — it must determine whether the system preserves the recognition window in foreseeable conditions of use.

6. Safeguards — Surrender-recognition capability is not an ethical add-on but the condition on which obligations that have bound parties for over a century remain effective. Its systematic exclusion engages existing duties and warrants documented justification.

X. CONCLUSION

The claim that fewer safeguards necessarily increase operational effectiveness or national security is incomplete. Certain safeguards function as operational mechanisms through which existing humanitarian obligations remain capable of being discharged — and their absence is a fact the law already has the instruments to examine.

The compliance challenges associated with autonomous systems are not solely questions of accuracy or capability. They are questions of time. Human judgment operates within biological limits, while computational systems increasingly operate at machine speed. Where this temporal asymmetry becomes sufficiently pronounced, meaningful human control becomes nominal rather than substantive, and compliance-relevant functions must be preserved within the architecture, because architectural safeguards may be among the few mechanisms operating within the window in which the decision is actually made.

Surrender recognition capability in operational AI systems is the paradigm case of such a lawful operational safeguard. The law does not need to be rewritten. Its existing protections must remain operative. No one can be required to do the impossible — *ad impossibilia nemo tenetur* — but an impossibility established at the design stage is not a circumstance encountered. It is a decision taken, and a decision can be examined.

Preserving a recognition window is one concrete way of ensuring that autonomous functions do not deprive a person of the practical possibility of surrendering, do not deprive the prohibition on attacking persons hors de combat of its practical operation, and do not deprive the prohibition binding on every party to an armed conflict of its intended effect.

If the analysis presented here is correct, the relevant question in future governance, procurement, and accountability discussions is not whether AI systems should have fewer or more safeguards as a general matter. It is which safeguards perform compliance-relevant functions under existing law, what verifiable criteria determine their technical feasibility, and how the sequence of technical feasibility → documented justification → command awareness → operational accountability structures legal review under existing obligations and, where applicable, formal legal-review duties.

METHODOLOGICAL NOTE

This paper proposes an analytical framework grounded in the customary prohibition examined in Section II and in the treaty provisions that state it for the States bound by them. It does not purport to resolve disputed questions of treaty interpretation or to establish binding standards of conduct beyond those imposed by the cited provisions themselves.

The qualifications previously distributed throughout the text are consolidated in this methodological note. The paper makes no claim that any specific technical implementation is legally mandated. The illustrative scenarios in Section V.D are analytical constructs and do not describe or evaluate any existing system or deployment. The temporal figures in Section IV are drawn from published literature and are presented as orders of magnitude illustrating a structural asymmetry, rather than as operational specifications. References to procurement frameworks and audit criteria are analytical reference points and do not constitute legal advice.

The analysis rests on customary international law, reflected in Rule 47 and Article 23(c) of the Hague Regulations. For States party to Additional Protocol I, Article 41 provides the clearest treaty formulation and Article 36 provides the procedural anchor where applicable. Comparable inquiries for non-party States rest on separate, expressly identified legal bases.

The author welcomes engagement, critique, and further development of the framework by scholars, practitioners, and policymakers working in international humanitarian law, autonomous systems governance, and defence procurement.³⁰

Cite as: Giovanni Nardacci, 'Lawful Operational Safeguards in AI Systems: Surrender Recognition, Compliance Architecture, and International Humanitarian Law' (6th rev. edn, 2026).

³⁰ All normative citations in this edition, customary and treaty alike, have been verified against the official texts (OHCHR instrument database; ICRC IHL databases) as of July 2026. They remain subject to confirmation by qualified counsel prior to formal submission.

Giovanni Nardacci is the founder of Human Flag Association, a Switzerland-based humanitarian technical initiative focused on civilian protection standards and IHL compliance architectures in AI-enabled operational environments. UNGM Partner No. 6126. humanflag.org. The author prepared this paper independently; no funding was received from any party with a commercial or governmental interest in the subject matter.

SELECT BIBLIOGRAPHY

Primary Sources — International Legal Instruments

Geneva Conventions of 12 August 1949, common Article 3.

Protocol Additional to the Geneva Conventions of 12 August 1949 (Protocol I), 8 June 1977, 1125 UNTS 3, Articles 36, 40, 41, 42, 57, 87.

Rome Statute of the International Criminal Court, 17 July 1998, Article 28.

Hague Convention IV respecting the Laws and Customs of War on Land (1907), Regulations, Art. 23(c).

Regulation (EU) 2024/1689 (EU Artificial Intelligence Act), Article 2(3), Annex III.

Jurisprudence

Legality of the Threat or Use of Nuclear Weapons (Advisory Opinion) [1996] ICJ Rep 226.

ICRC, UNIDIR and Official Materials

Henckaerts, J.-M. & Doswald-Beck, L., Customary International Humanitarian Law, Vol. I: Rules (ICRC/CUP, 2005), Rules 15–16, 46–47.

ICRC, A Guide to the Legal Review of New Weapons, Means and Methods of Warfare (Lawand ed., 2006).

ICRC, Interpretive Guidance on the Notion of Direct Participation in Hostilities under IHL (2009).

ICRC, Autonomous Weapon Systems and IHL: A Guide to the Issues (2014); Autonomous Weapons Systems: Implications of Increasing Autonomy in the Critical Functions of Weapons (2023).

Pictet, J. S. (ed.), *Commentary on the Geneva Conventions of 12 August 1949, Vol. IV* (ICRC, 1958).

Sandoz, Y., Swinarski, C. & Zimmermann, B. (eds.), *Commentary on the Additional Protocols of 8 June 1977* (ICRC/Martinus Nijhoff, 1987).

UNIDIR, *The Interpretation and Application of International Humanitarian Law in Relation to Lethal Autonomous Weapon Systems* (2025).

US Department of Defense, *Law of War Manual* (July 2023 update), §§ 5.4.3, 5.5.3; Directive 3000.09 (updated 2023).

Academic and Technical Literature

Carlini, N. & Wagner, D., 'Towards Evaluating the Robustness of Neural Networks' (2017) *IEEE S&P* 39.

Goodfellow, I. J., Shlens, J. & Szegedy, C., 'Explaining and Harnessing Adversarial Examples' (2015) *ICLR*.

Heusler, B. & Sutter, C., 'Shoot or Don't Shoot?' (2022) 13 *Frontiers in Psychology* 798766.

Hick, W. E., 'On the Rate of Gain of Information' (1952) 4 *Quarterly Journal of Experimental Psychology* 11.

Kosinski, R. J., 'A Literature Review on Reaction Time' (Clemson University, updated 2013).

Nardacci, G., 'Systemic Arrestability: Operator Protection in Machine Safety Law When Machine Speed Exceeds Human Reaction' (Human Flag Association working paper, 2026), Zenodo DOI 10.5281/zenodo.20837150.

Payne, K., I, *Warbot: The Dawn of Artificially Intelligent Conflict* (Hurst, 2021).

Ratcliff, R. & McKoon, G., 'The Diffusion Decision Model' (2008) 20 *Neural Computation* 873.

Sassoli, M., 'Autonomous Weapons and International Humanitarian Law' (2014) 90 *International Law Studies* 308.

Scharre, P., *Army of None: Autonomous Weapons and the Future of War* (W.W. Norton, 2018).

Schmitt, M. N. & Thurnher, J. S., 'Out of the Loop' (2013) 4 Harvard National Security Journal 231.

Szegedy, C. et al., 'Intriguing Properties of Neural Networks' (2014) ICLR.

Policy and Institutional Documents

Article 36, Killer Robots: UK Government Policy on Fully Autonomous Weapons (2013).

Human Rights Watch & IHRC, Losing Humanity (2012); Mind the Gap (2015).

IEEE 7000-2021; NATO STO, Technology Readiness Levels in NATO Science and Technology (2019); NIST, AI Risk Management Framework (AI RMF 1.0, 2023).

RAND Corporation, Human-Machine Teaming for Future Ground Forces (2020); SIPRI, AI in Military Systems and its Implications for the Arms Control Regime (2021).

UK Ministry of Defence, Ambitious, Safe, Responsible (2022); UNIDIR, The Weaponization of Increasingly Autonomous Technologies (2014).